



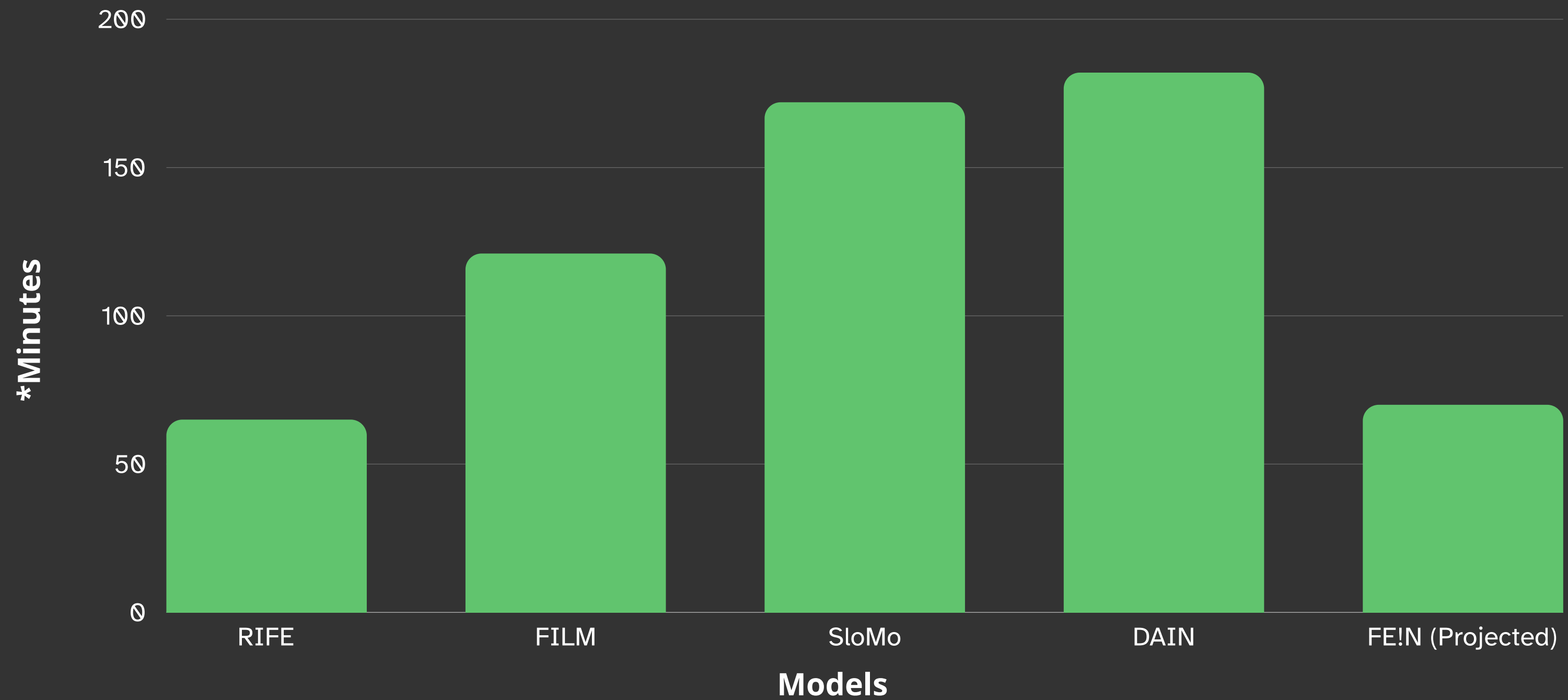
FRAME INTERPOLATION GIST



CHOICE OF MODELS

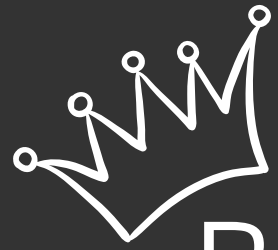
- FILM: Frame Interpolation for Large Motion
- DAIN: Depth-Aware Video Frame Interpolation
- Super SloMo (Pytorch): Super Slow Motion
- RIFE: Real-Time Intermediate Flow Estimation

Interpolation Performance (5)



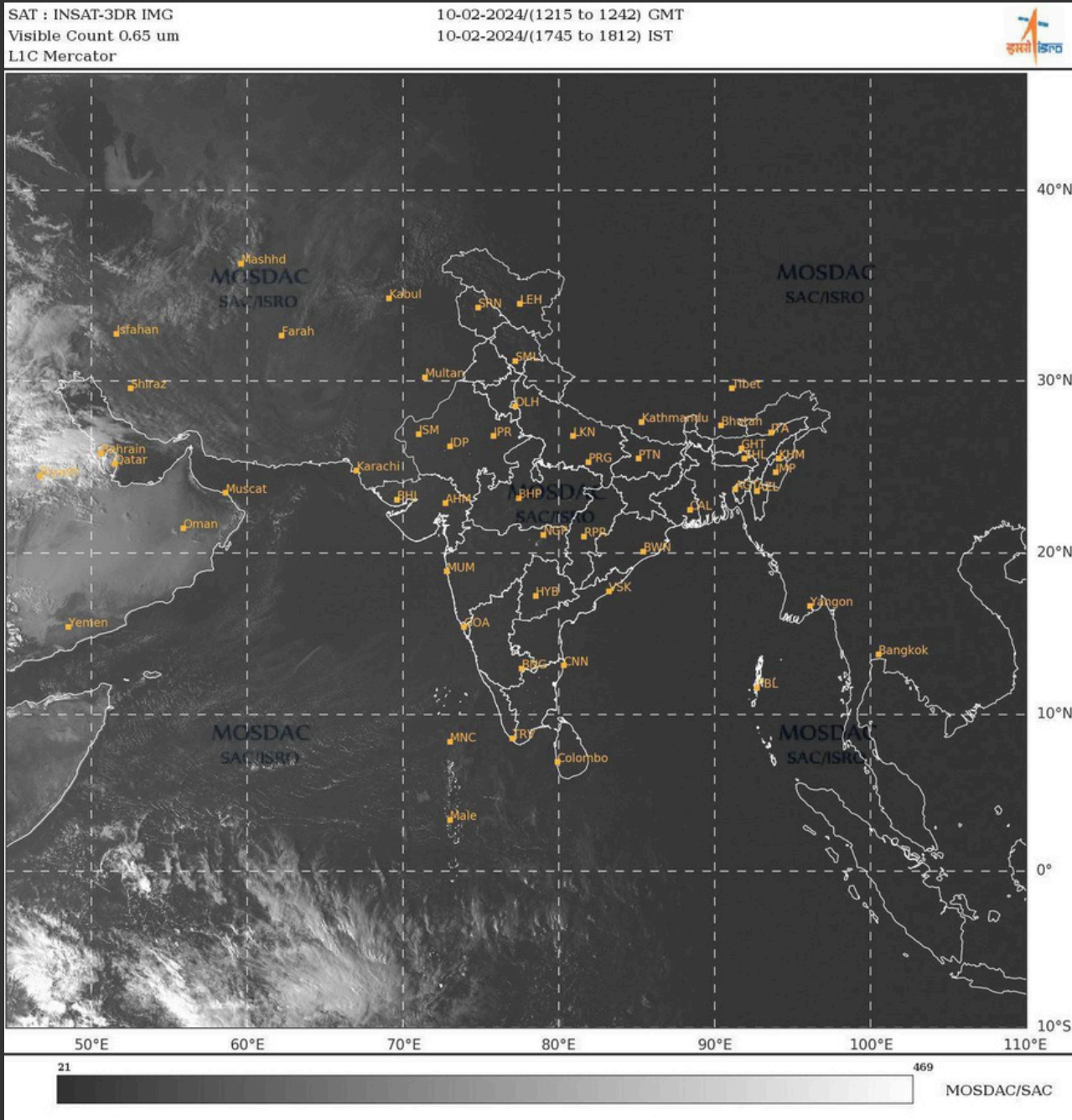
*RTX 3050 Intel i7-12650H

VERDICT

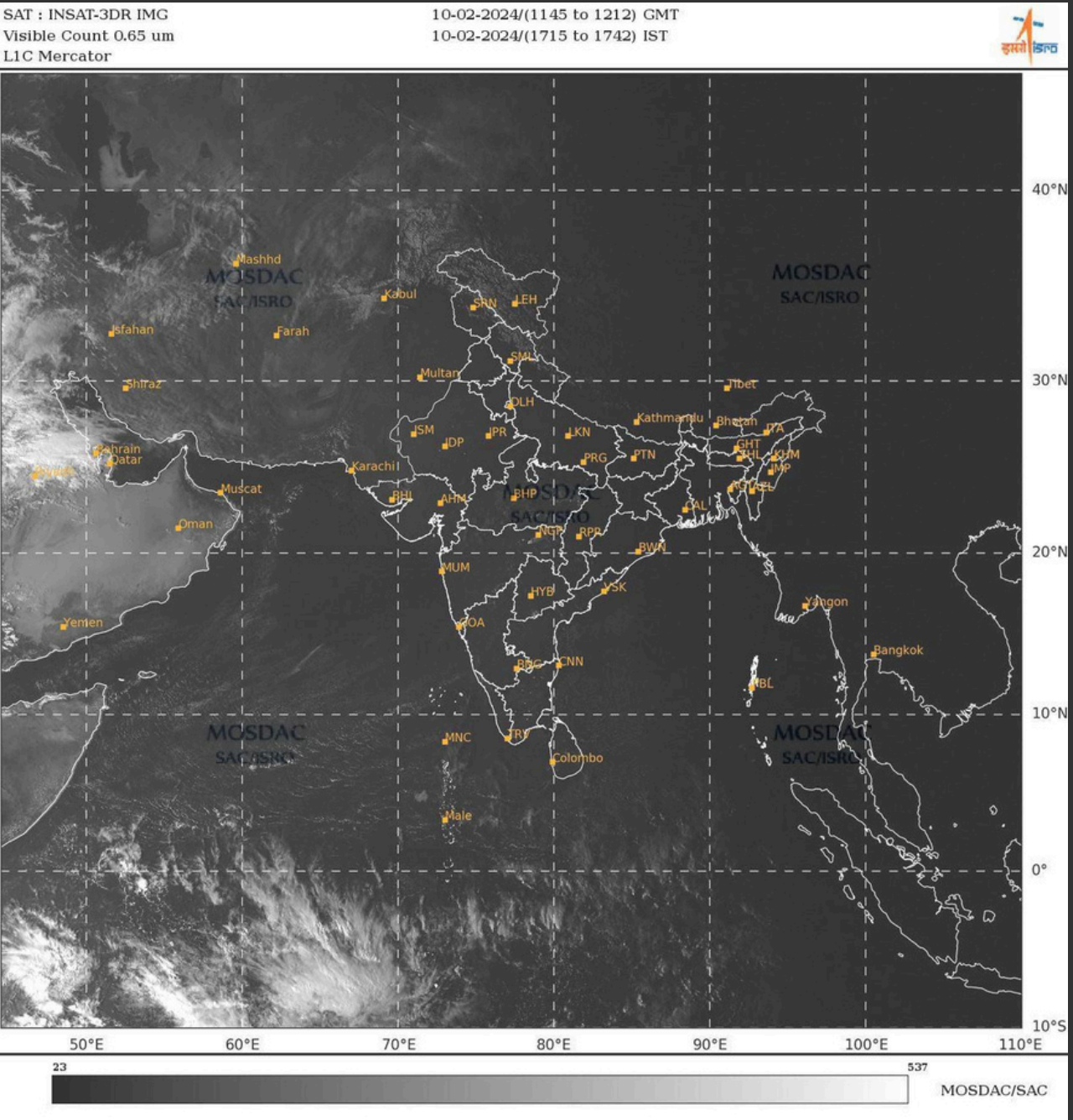


RIFE is the best (our usecase) at Frame interpolation taking advantage of it's **streamlined architecture, optical flow estimation, bidirectional flow interpolation and perceptual loss**

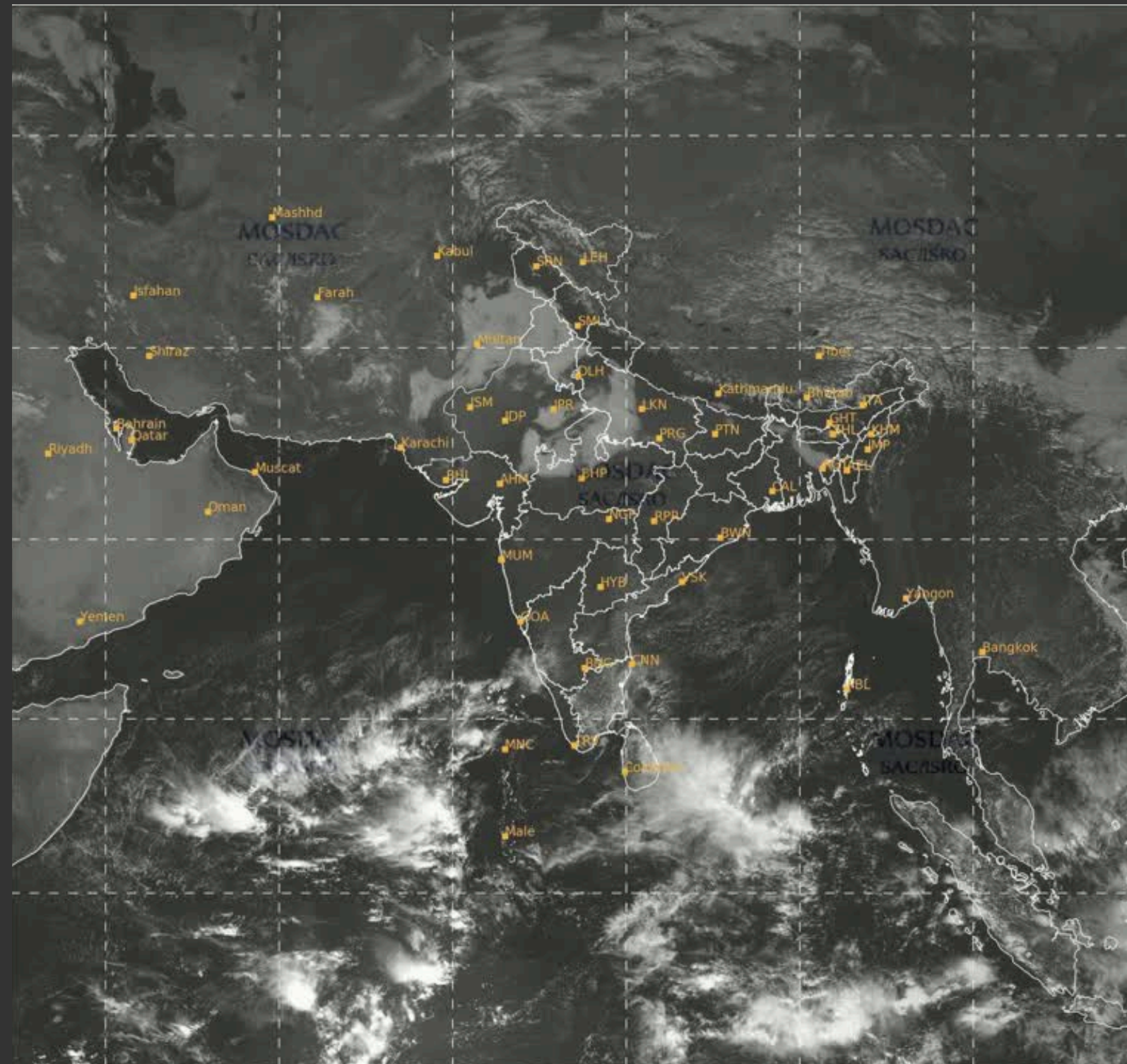
RIFE



Generate
Intermediate Video



VIDEO



RIFE NOT ENOUGH

- Not Specific for Satellite Data
- Does not Consider Natural Movements like Winds
- Not Refined enough for the use case we are tackling
- Complicated Computation undertaken for known values (Direction Vectors)
- No cloud interpolation at night

FE!N

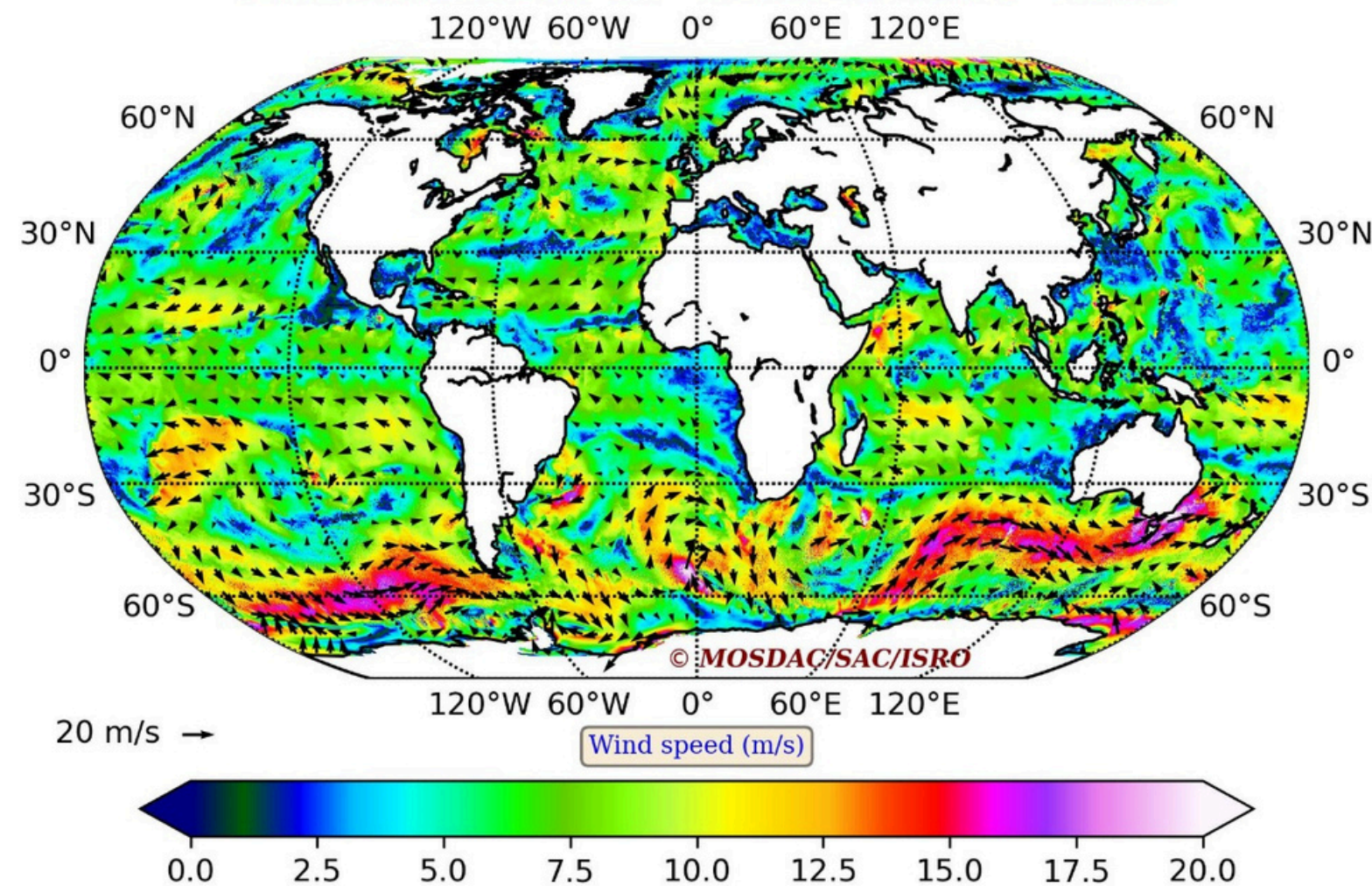
- FE!N – Flow Enhanced Interpolation Network
- Our Custom Model which will be specific to the video we generate
- Will incorporate all the Calibrations and necessary modifications to go about the video generation

FE!N

- FE!N takes in data from MOSDAC (EOS-06 WIND)
- Distributes data in Vectors $\hat{i}$ & $\hat{j}$ converts it into feature map
- Computes it with Cloud Mask Extraction
- Encode
- Generates the Loss Function
- Output Generated

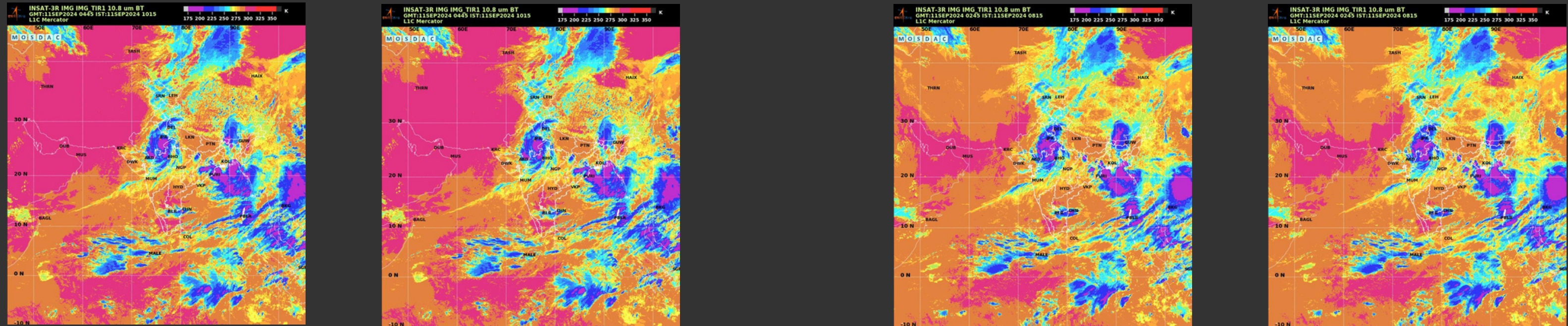
EOS-06

EOS-06 scatterometer Level-4AW Analyzed Wind
Dated: 2024-09-02 12:00:00 GMT



- EOS-06 Data
- Updates every 24 hours
- Has directional data for Wind
- Converting this data into Matrices for further computation

DATASET

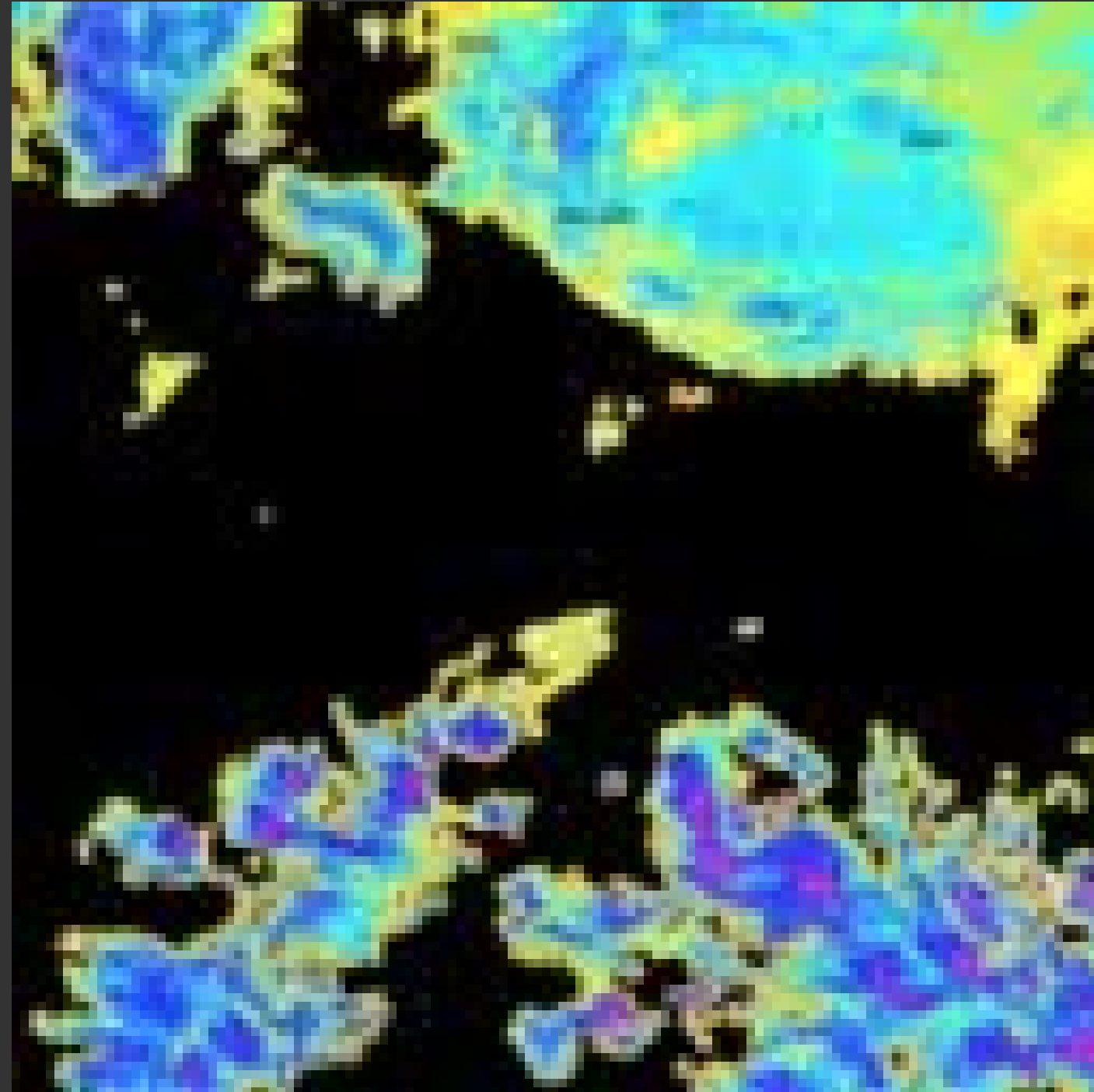


- 1500+ Images spanning over 1 month of INSAT 3DR data

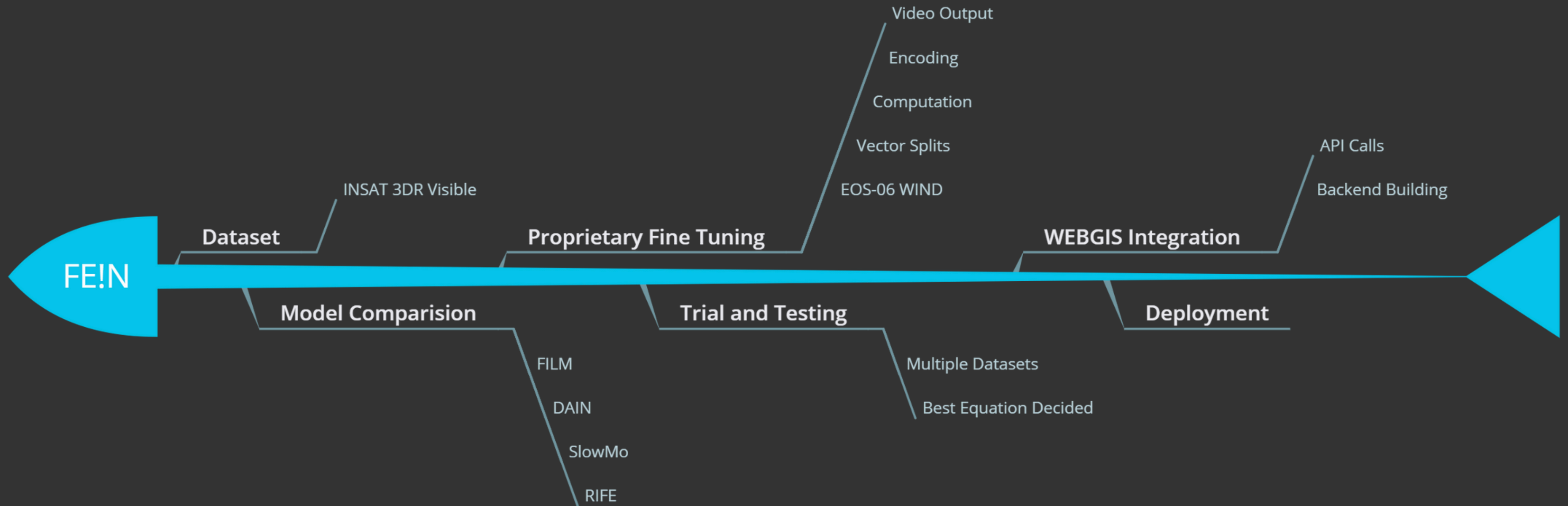
WHY FE!N

- Promised Better Precision as more factors taken in account
- Reduced Computation – Direction Vectors Simplified
- Proprietary Training – Maxxed out Accuracy
- Trial and Testing – INSAT 3DR data exclusively used
- Can predict clouds at night

OUTPUT



TIMELINE



THANK YOU